AI211L	Artificial Intelligence and Machine Learning (1 CH)	Knowledge Profile: WK3	Focus: CCP	CS/AI

Prerequisite: CS101

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Office Hours: 9:00 am ~ 2:30 pm (Mon -Wed)

Lab Introduction

This lab introduces the practical foundations of Artificial Intelligence by emphasizing the implementation of core AI concepts in Python. Students will work with intelligent agents, search algorithms, game strategies, constraint satisfaction problems, optimization techniques, and machine learning. The focus is on solving real-world problems through algorithmic design, coding, and experimentation. By the end of this lab, students will be able to apply AI techniques to model decision-making, implement supervised and unsupervised learning algorithms, and evaluate intelligent systems using modern libraries such as Scikit-learn, TensorFlow, and NetworkX.

Lab Contents

This lab will cover the following topics:

1. AI Foundations and Intelligent Agents:
Introduction to Artificial Intelligence and basic Python programming concepts; agents and environments; intelligent agents; simple reflex agents and model-based reflex agents; basic decision-making and problem-solving using Python.

2. Search and Problem Solving:
Problem formulation and state-space representation; uninformed search techniques including Breadth-First Search (BFS) and Depth-First Search (DFS); heuristic search; Greedy Best-First Search; A* algorithm; pathfinding and cost functions.

3. Local Search and Optimization:
Introduction to local search and optimization; objective and fitness functions; Hill-Climbing search; Simulated Annealing; Genetic Algorithms; population generation, fitness evaluation, selection, crossover, and mutation.

4. Constraint Satisfaction and Adversarial Search:
Constraint Satisfaction Problems (CSPs); variables, domains, and constraints; problem solving using CSPs; adversarial search; two-player games; Minimax algorithm and Alpha-Beta pruning.

5. Knowledge Representation, Logic and Probabilistic Reasoning:
Knowledge representation; propositional logic; logical inference; basic First-Order Logic (FOL); representation of facts and rules; probabilistic reasoning; Bayes' theorem and basic Bayesian reasoning.

6. Supervised Machine Learning:
Introduction to machine learning; Linear Regression; Logistic Regression; classification using K-Nearest Neighbors (KNN) and Decision Trees; training and testing; model evaluation; generalization; overfitting and underfitting.

7. Unsupervised Learning:
Introduction to unsupervised learning; similarity and distance measures; clustering techniques; K-Means clustering; determining the number of clusters; cluster evaluation and interpretation.

8. Dimensionality Reduction:
Introduction to dimensionality reduction; Principal Component Analysis (PCA); feature transformation; explained variance; selection of principal components; visualization and reduced-dimensional representations.

Mapping of CLOs and PLOs

Sr. No.	Course Learning Outcomes (CLOs)[+]	Graduate Attributes (GAs)	Bloom's Taxonomy level (P Domain)
CLO 1	***Apply*** core AI concepts such as intelligent agents and searching techniques using Python.	GA-2 (Knowledge for Solving Computing Problems)	P3 – Guided Response
CLO 2	***Analyze*** and design solutions for real-world problems using state-space search, CSPs, and adversarial strategies.	GA-4 (Design/Development of Solutions)	P4- Mechanism
CLO 3	Develop and ***evaluate*** supervised and unsupervised learning algorithms using modern tools (Scikit-learn, TensorFlow).	GA-5 (Modern Tool Usage)	P4- Mechanism
	[+]Please add the prefix "Upon successful completion of this course, students will be able to"		

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Lab Performance	30%	30%	20%
OEL/Project	20%	20%	40%
Midterm Exam	50%	50%	
Final Exam			40%

Overall Grading Policy

Assessment Items	Percentage
Lab Performance	40%
OEL/Project	10%
Midterm Exam	20%
Final Exam	30%

Text and Reference Books

Textbook:

- Artificial Intelligence – A modern approach, 4th edition, by Stuart Russell and Peter Norvig (2021)
- Artificial Intelligence – Latest Advances, New Paradigms and Novel Applications, 5th edition by Eneko Osaba, Esther Villar, Jesus L. Lobo and Ibai Lana (2021)
- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd edition by Aurelien Geron

Reference books:

- Artificial Intelligence in the 21st Century by Stephan Lucci and Danny Kopek (2012)

Administrative Instruction

- According to institute policy, 80% attendance is mandatory to appear in the final examination.
- Attendance is mandatory for students in all the labs. If a student is absent from a lab due to any reason, he/she will have to get written permission of the Dean to perform that lab. The Dean will allow students to perform lab if he feels that the student has agenuine excuse.
- Students should bring their textbooks to the lab, so that they can refer to theory and diagrams whenever required. `
- Labs will be graded in double-entry fashion; one entry in the assessment sheet given at the end of every lab and another entry inthe instructor's record. This system of keeping records will keep students aware of their performance throughout the lab.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Computer Usage/Software Tool

- Python, NetworkX, Scikit-learn, Tensorflow, VScode, Jupyter Notebook, Google Colab

Weekly Breakdown

Week 1	• Introduction to AI with Python, Python Libraries
Week 2	• Intelligent Agents (Simple & Model Based Reflex Agents).
Week 3	• Uninformed Search — BFS and DFS • Informed Search — Heuristics and Greedy Best-First Search • A* Algorithm
Week 4	• Local Search and Optimization • Hill-Climbing • Simulated Annealing • Genetic Algorithms
Week 5	• Constrained Satisfaction Problem (CSPs)
Week 6	• Adversarial Search • Minimax Algorithm • Alpha Beta Pruning
Week 7	• Knowledge Representation, Logic & Probabilistic Reasoning
Week 8	• Machine Learning — Linear & Logistic Regression
Week 9	• Classification — KNN, Decision Trees & Model Evaluation
Week 10	• Unsupervised Learning — K-Means
Week 11	• Dimensionality Reduction — Principal Component Analysis (PCA)
Week 12	• OEL / Project